THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapters 5–8: **Adolescence: A Stage of Growth and Change, Heat Transfer in Nature, Measurement of Time and Motion, Life Processes in Animals** | Class 7 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(b)** cyberbullying	avoid and report
2	**(d)** Using sanitary pads and disposing them properly	health and cleanliness
3	**(a)** Iron	spinach, raisins
4	**(c)** trap air, a poor conductor	air between fibres
5	**(a)** air trapped between them	insulation by air
6	**(a)** degree Celsius	°C
7	**(c)** 30 m/s	1 km/h = 5/18 m/s
8	**(b)** 120 km	distance = speed × time
9	**(a)** 4 hours	time = distance ÷ speed
10	**(a)** surface area for absorption	villi
11	**(c)** digestion	in the alimentary canal
12	**(c)** respiration	in body cells

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	The stage when the body develops to be capable of reproduction (2).	2
14	They are poor conductors and do not burn hands (2).	2
15	They reflect more heat (2).	2
16	72 × 5/18 = 20 m/s (2).	2
17	They have no teeth; the gizzard with grit grinds food (2).	2

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	Causes blood-related problems (1.5); Dorothy Hodgkin (1.5).	1½+1½
19	Falling water table (1.5); rainwater harvesting and afforestation (1.5).	1½+1½
20	5 hours (3).	3
21	Bile and pancreatic juice neutralise acid (1); intestinal juice completes digestion (1); villi absorb nutrients (1).	1+1+1
22	Inhalation: ribs up, diaphragm down, air in (1.5); exhalation: ribs down, diaphragm up, air out (1.5).	1½+1½
	OR: Oxygen 21% vs 16–17% (1.5); CO_2 0.04% vs 4–5% (1.5).	1½+1½
23	Rapid growth needs energy (1); proteins from pulses, calcium from milk, iron from spinach (2).	1+2
	OR: Improve adolescent health (1); peer education (1); awareness about menstrual health (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	(i) Falling water table (1). (ii) Rainwater harvesting (1). (iii) Seepage of water (1). (iv) Check dams or percolation pits (2).	1+1+1+2
	OR: (i) Colour matters (1). (ii) Dark absorbs, light reflects (2). (iii) White in summer, dark in winter (2).	1+2+2
25	(i) Nutrients enter blood (1). (ii) Finger-like projections (2). (iii) Carried to cells for energy and growth (2).	1+2+2
	OR: (i) Ingestion, digestion, absorption, assimilation, egestion (1). (ii) Stepwise (3). (iii) Bile; pancreatic juice (1).	1+3+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapters 5–8: **Adolescence: A Stage of Growth and Change, Heat Transfer in Nature, Measurement of Time and Motion, Life Processes in Animals** | Class 7 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(c)** talk to trusted elders and take part in sports or music	positive growth
2	**(b)** milk	bone growth
3	**(b)** the influence of friends on one's behaviour	adolescents mimic peers
4	**(b)** evaporation	Sun's heat
5	**(d)** 35 °C and 42 °C	body temperature range
6	**(a)** aquifers	pore spaces
7	**(b)** distance covered per unit time	definition
8	**(b)** 36 km/h	multiply by 18/5
9	**(d)** regular oscillation of a pendulum	to measure time
10	**(b)** small intestine	villi
11	**(d)** gastric juice (with acid)	acid plus enzymes
12	**(b)** downwards	chest space increases

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	Oily secretions clog skin pores (1); wash face, balanced diet, avoid picking (1).	1+1
14	Heat transfer by movement of a fluid (1); boiling water or sea breeze (1).	1+1
15	150 ÷ 3 = 50 km/h (2).	2
16	Distance travelled per unit time (1); speed = distance ÷ time (1).	1+1
17	Both are basic and neutralise acid (1); bile breaks fats, pancreatic juice digests all food types (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	Be respectful, think before posting (1.5); do not share personal photos or passwords (1.5).	1½+1½
19	Heated air is less dense and rises by convection (3).	3
20	25 km (1.5); 25/3 = 8.33 km/h (1.5).	1½+1½
21	Chewing and saliva begin starch digestion (1.5); stomach acid and enzymes digest proteins (1.5).	1½+1½
22	Height and weight increase (1); voice changes, hair growth (1); pimples and body shape changes (1).	1+1+1
	OR: It is a natural process (1.5); no scientific basis for taboos (1.5).	1½+1½
23	White paint, ventilation, shades (3).	3
	OR: Stores water underground for wells and borewells (3).	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	(i) Mood swings, strong feelings (2). (ii) Sensitivity, risk-taking (1). (iii) Hobbies, talking, exercise (2).	2+1+2
	OR: (i) Proteins, carbohydrates, fats, vitamins, minerals (1). (ii) Spinach; milk (2). (iii) Rapid growth (1). (iv) RKSK (1).	1+2+1+1
25	(i) Sun shadow; water flow (3). (ii) Sundial needs sun; clepsydra needs refilling (2).	3+2
	OR: (i) 8 m/s (1). (ii) 28.8 km/h (1). (iii) 50 s (1). (iv) Stopwatch (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET C

Chapters 5–8: **Adolescence: A Stage of Growth and Change, Heat Transfer in Nature, Measurement of Time and Motion, Life Processes in Animals** | Class 7 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(c)** building stamina and improving mood	healthy body and mind
2	**(b)** Dorothy Hodgkin	third woman Nobel in chemistry
3	**(a)** secondary sexual characteristics	e.g. facial hair, breast development
4	**(d)** A black cloth	dark absorbs
5	**(d)** water cycle	evaporation, condensation, rain
6	**(a)** hot air rises	convection
7	**(a)** non-uniform motion	changing speed
8	**(b)** second	s
9	**(c)** 10 m/s	1 km/h = 5/18 m/s
10	**(c)** rectum	lower part of large intestine
11	**(a)** spiracles	tracheal system
12	**(b)** diaphragm	model

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	A monthly cycle in girls (1); 28–30 days, may vary 21–35 (1).	1+1
14	Cool wind from the sea to the land during the day (2).	2
15	Equal distances in equal times; unequal distances in equal times (1+1).	1+1
16	Increases surface area for absorption (2).	2
17	Removal of undigested waste as stool through the anus (2).	2

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	Bump formed by growth of the voice box (1.5); more prominent in boys (1.5).	1½+1½
19	Solids/medium contact (1); fluids movement (1); no medium (1).	1+1+1
20	25 m/s (1.5); 28.8 km/h (1.5).	1½+1½
21	Oxygen passes into blood (1.5); carbon dioxide released into alveoli (1.5).	1½+1½
22	Vacuum prevents conduction/convection (1.5); silvered walls reduce radiation (1.5).	1½+1½
	OR: Day: land hotter, hot air rises, cool wind from sea (1.5); night: reverse (1.5).	1½+1½
23	Hang a bob and swing small angle (1); time 20 oscillations with a stopwatch (1); divide to get time period (1).	1+1+1
	OR: Distance is how far; speed is how fast (1.5); units m and m/s (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	(i) Heat passes easily (1). (ii) Metals; wood, plastic (2). (iii) Handles, wool, thermos (2).	1+2+2
	OR: (i) Sea and land breeze (1). (ii) Land hotter (1). (iii) Sea warmer (1). (iv) Fishermen and sailors use them (2).	1+1+1+2
25	(i) Physical vs chemical (2). (ii) Body cells (1). (iii) Energy (2).	2+1+2
	OR: (i) Rumination (1). (ii) Digest grass (1). (iii) Rumen (1). (iv) Gizzard grinds (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.